Active Maintenance Report: schwarz_coppock_2022

Schwarz and Coppock (2022, Journal of Politics): What Have We Learned about Gender from Candidate Choice Experiments?

2026-08-01

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Drafted by Claude Opus 5 under the supervision of Alex Coppock.

This repository holds the actively maintained replication code for Schwarz and Coppock (2022), together with the reproducibility report that documents what the original archive did and did not do. It is part of a program applying the maintenance proposal in Peer, Orr and Coppock (2021, *PS: Political Science & Politics*, doi 10.1017/S1049096521000366) to a set of published archives.

Article	10.1086/716290
Replication archive	10.7910/DVN/R27ULT

The data are not redistributed here. The deposit is 0.5 MB across 12 files and lives at Harvard Dataverse, which is the only copy this repository points at. `download_original.R` fetches it and verifies every file; `original_manifest.csv` pins the file identifiers, sizes and checksums, so the exact bytes this code was written against are recorded in version control even though the bytes themselves are not.

Repository layout. `maintained/` is the maintained rewrite: one script per published table or figure, writing to `output/`, which is committed so a reader can compare a fresh run against it without downloading anything. `ground_truth/` ties every published number to the code that produces it. `original/` is created by the download script and is deliberately absent from the repository. This file is the reproducibility report, also available as a PDF in `report/`.

License. CC0 1.0 Universal, matching the terms of the deposit this repository maintains. See `LICENSE`.

To reproduce. Clone or download the repository, open `schwarz_coppock_2022.Rproj`, and run:

```
source("run_all.R")
```

That fetches the deposit, verifies its 12 files, and produces every table and figure into `maintained/output/`. Required packages: tidyverse, metafor, knitr, kableExtra, here. Paths resolve through `here`, so nothing depends on the working directory. The full run takes about six seconds. A successful run overwrites `maintained/output/`, which is committed: **`git diff` on that folder is the reproduction check.**

1 Summary

Two questions, answered before the detail.

1.1 Does the deposited archive run?

Yes, completely. All five analysis scripts execute without error on a current R installation. There are no `setwd()` calls, no hardcoded absolute paths and no functions called before they are defined, and every package the scripts name still installs from CRAN, `rmeta` and `ggstance` included. Nothing in the deposit needs a substitution to run.

What the archive does not do is write anything. All eight `ggsave()` calls, all three `print.xtable()` calls and the single `sink()` block are commented out in the deposited files, so a clean run leaves nothing behind but a stray `Rplots.pdf` from the plots being printed to a device. There is nothing to diff a rerun against, which is why the maintained rewrite writes its outputs to disk.

1.2 Does the maintained rewrite reproduce the paper?

Mostly, and the exceptions are in the article rather than in the code. 72 of the 78 verifiable ground truth claims match the published values to reported precision, including every pooled estimate and standard error printed on Figures 1 through 4 and all forty cells of Table 2. The remaining 22 recorded quantities are ones the article never prints, most of them confidence interval bounds that the forest plots draw but do not label.

The 6 failures are all counts stated in prose, and none of them changes a substantive conclusion:

- The article reports that of the 67 estimates, "47 are positive (23 significant) and 18 are negative (11 significant)." The deposit has 47 positive, **19** negative and one exactly zero, which is the only split that sums to 67; 47 plus 18 sums to 65. Neither of the two plausible significance rules recovers 23 and 11. The rule the archive's own script applies, a confidence interval excluding zero, gives 29 positive-and-significant and 6 negative-and-significant. Testing p below 0.05 instead gives 19 and 2, and cannot be what was done in any case, because 19 of the 67 rows carry no p-value at all.
- The article describes Figure 3 as covering "the 36 experiments where we were able to identify respondent gender." The deposit has 37, and so does the published figure, whose panel A prints 37 study rows.
- The discussion says "more than three-fourths of the studies show a positive effect." The share is 47 of 67, or 70.1%.
- Table A.1 lists 43 studies where the appendix carries 44 per-study CATE figures; Simas and Murdoch (2019) has Figure A.44 but no Table A.1 row.

Every estimate, standard error, interval and p-value the article reports reproduces exactly. What does not reproduce is a set of hand-counted descriptions of those estimates.

2 Paper overview

Citation: Schwarz, S. and Coppock, A. (2022). "What Have We Learned about Gender from Candidate Choice Experiments? A Meta-Analysis of Sixty-Seven Factorial Survey Experiments." *The Journal of Politics*, 84(2), 655-668. DOI: 10.1086/716290

Summary: Sixty-seven conjoint and factorial candidate choice survey experiments, fielded between 1982 and 2021 in more than twenty countries, all randomized candidate gender

alongside other candidate attributes. Each study's treatment effect of being described as a woman is re-estimated on a common footing and the 67 estimates are pooled by random-effects meta-analysis. The pooled effect is a gain of 1.8 percentage points of vote share for women candidates, with a confidence interval from 1.1 to 2.5 points. The effect is larger for white than for Black candidates in the US context (2.2 against 0.9 points), larger among women respondents than men (3.0 against 0.9), and reverses sign among Republicans (-1.4) while running positive among Democrats (4.2) and Independents (3.3).

What the deposit contains: five pre-computed summary datasets and five analysis scripts. The individual-level data belong to the authors of the 67 original studies and are not redistributed, so the deposit begins one level up, at estimated ATEs and CATEs with their standard errors. Every number in the article is a function of those five files.

3 Original archive reproducibility

Table 1: The deposited scripts on a current R installation.

Script	Status on current R	Resolution
helpers.R	Clean (sourced by all five scripts)	No changes required
meta_analyze_sates.R	Runs; writes nothing, every output call is commented out	No changes required
meta_analyze_cates_cand_race.R	Runs; writes nothing	No changes required
meta_analyze_cates_resp_gender.R	Runs; writes nothing	No changes required
meta_analyze_cates_resp_pid.R	Runs; writes nothing. Reports the Independent-Republican p-value from the Democrat-Republican statistic	Corrected in the rewrite; see Errata
meta_analyze_cates_cand_attributes.R	Runs; writes nothing. Its per-study figure loop is written but its writes are commented out	Restored in the rewrite as 44 written figures

Nothing here needs fixing to make the archive run. The deprecations are all warnings rather than errors: `geom_errorbarh()` and the `%+%` operator are deprecated in ggplot2 4.0.0 but still work, `reshape2::melt()` and `dcast()` are superseded, and `dplyr::do()` has been superseded since 1.0.0. What is wrong with this archive is the p-value bug documented in the Errata, which is a logic error rather than a run failure.

`rmeta` and `ggstance` are both still installable. The maintained rewrite replaces `rmeta::meta.summaries()` with `metafor::rma()` on maintenance grounds rather than availability grounds: `rmeta` has not been updated since March 2018. The two implementations of DerSimonian-Laird agree on every pooled estimate, standard error and interval bound in the

paper to more than ten significant figures. `ggstance::position_dodgev()` resolves without incident from ggstance 0.3.7, which `coefplot` pulls in. It is dropped from the rewrite for a different reason: it is inert. Building the layer with and without it puts every point at the same coordinates, because every y value it is asked to dodge is already unique, 954 distinct labels across 954 rows in Figure A.1 and no repeated label within any study in Figures A.2 through A.45.

What the archive does not produce. Every write in the deposit is commented out: eight `ggsave()` calls, three `print.xtable()` calls and one `sink()` block. Two of the commented `ggsave()` paths point at a directory that was never part of the deposit (`04_Drafts/current_draft/figures/`). The consequence is that the deposit cannot be checked against itself. There is no committed output to diff a rerun against, and a reader who runs all five scripts sees console output and a `Rplots.pdf` and nothing else. Restoring those writes, so that a rerun can be compared against a committed result, is the single largest change the maintained rewrite makes.

4 Errata

The Independent-versus-Republican p-value is computed from the wrong statistic. In `meta_analyze_cates_resp_pid.R`, the block that tests the two differences between pooled partisan CATEs computes the second p-value from the first test's statistic:

```
# Archive:
I_R_statistic = I_R_estimate / I_R_std.error,
I_R_p.value = 2 * pnorm(abs(D_R_statistic), lower.tail = FALSE)

# Rewrite:
I_R_p.value = 2 * pnorm(abs(I_R_statistic), lower.tail = FALSE)
```

`I_R_statistic` is computed on the line above and then never used, so the reported Independent-Republican p-value is numerically identical to the Democrat-Republican one.

Quantity	Published	Original archive	Maintained rewrite
Independent minus Republican, p	not reported	2.21e-10 (equals the D-R p-value)	1.06e-07

The article states only that "the Republican estimate is statistically significantly different from the Democratic and Independent estimates," which is true of both the wrong value and the right one. No published number changes.

Not corrected: the trimming rule. Table 2's last-but-one row is labelled "Trimmed (middle 90% of estimates)" and the text says the row drops "the bottom and top 5% of point estimates." The code keeps estimates above the 2nd percentile and below the 95th, retaining 61 of 67 studies. Correcting 0.02 to 0.05 would be a plausible reading of a typo, but it is a change to a sample definition rather than to a mechanical error, so the rewrite reproduces the published row as computed and reports the alternative separately in `output/table_2_trim_sensitivity.csv`. The symmetric rule the text describes gives 0.020 over 59 studies against the published 0.018 over 61, so the robustness claim the row is making survives either reading.

Not corrected: the year threshold. The subset labelled "Post-2014 studies" is selected by `year > 2010`. The labels are nonetheless accurate, because the sample contains no study fielded between 1998 and 2014, a gap the article notes in its fourth footnote, so every threshold in that window produces the same two groups of 63 and 4.

5 Ground truth

Table 2: Ground truth: 100 recorded quantities, by where the article states them.

Table or figure	Reproduces	Not stated in the article	Does not reproduce
figure_1	5	0	0
figure_2	6	5	0
figure_3	6	5	1
figure_4	7	8	0
figure_a1	3	0	0
table_1	3	1	0
table_2	40	1	0
table_a1	0	0	1
text	2	2	4

`ground_truth/schwarz_coppock_2022_ground_truth.csv` records 100 quantities. `value_script` is what the deposited scripts produce through `rmeta`, `value_rewrite` is what `maintained/` produces through `metafor`, and `value_paper` is filled only by reading the article: from Table 1 and Table 2, from the article's prose, and from the pooled-estimate labels printed inside Figures 1 through 4, which are rasterized in the published PDF and were read off page renderings. Where the article does not state a quantity the cell is empty and `match` is `NA`, which is the case for every confidence interval bound in the four figures: the forest plots draw the intervals but label only the estimate and its standard error.

Table 3: The six published quantities that do not reproduce.

Table or figure	Claim	Article	Deposit	Note
text	n_negative	18.00	19.000	Text: 18 are negative. The deposit has 19 negative, one exactly zero and 47 positive, which sums to 67; the published 47 plus 18 sums to 65
text	n_pos_and_sig_ci_rule	23.00	29.000	Text: 47 estimates are positive (23 significant). The archive script's rule, a confidence interval excluding zero, gives 29
text	n_neg_and_sig_ci_rule	11.00	6.000	Text: 18 are negative (11 significant). The archive script's rule gives 6
text	prop_positive	0.75	0.701	Discussion states more than three-fourths of the studies show a positive effect. value_paper records that lower bound rather than an equality; 47 of 67 is 0.701, so the claim does not hold
figure_3	n_studies_in_text	36.00	37.000	Text: for the 36 experiments where we were able to identify respondent gender. The deposit and the paper's own figure both have 37
table_a1	n_studies_listed	43.00	44.000	Published Table A.1 lists 43 studies. Simas and Murdoch (2019) has a CATE figure at A.44 but no Table A.1 row

6 Maintained rewrite

Table 4: The maintained rewrite.

Script	Output
helpers.R	Packages, re_meta(), diff_test(), label formatting
table_1_study_manifest.R	table_1_study_manifest.csv
table_2_meta_robustness.R	table_2_meta_robustness.csv, table_2_trim_sensitivity.csv
table_a1_study_attributes.R	table_a1_study_attributes.csv
figure_1_sate_forest_plot.R	figure_1_sate_forest_plot.pdf and .png, figure_1_meta_summary.csv

figure_2_cate_candidate_race.R	figure_2_cate_candidate_race.pdf and .png, figure_2_meta_summary.csv, figure_2_diff_in_cates_cand_race.csv
figure_3_cate_respondent_gender.R	figure_3_cate_respondent_gender.pdf and .png, figure_3_meta_summary.csv, figure_3_diff_in_cates_resp_gender.csv
figure_4_cate_respondent_pid.R	figure_4_cate_respondent_pid.pdf and .png, figure_4_meta_summary.csv, figure_4_diff_in_cates_resp_pid.csv
figure_a1_cate_candidate_attributes.R	figure_a1_cate_candidate_attributes.pdf and .png, figure_a1_cate_summary.csv
figure_a2_a45_cate_by_study.R	44 per-study appendix figures, figure_a2_a45_index.csv
text_in_text_numbers.R	text_in_text_numbers.csv

Substitutions.

Deposited	Maintained	Why
`rmeta::meta.summaries(method = "random")`	`metafor::rma(method = "DL")`	Same DerSimonian-Laird estimator; `rmeta` last updated 2018
`do(tidy(with(., ...)))`	`group_by() |> reframe(re_meta(...))`	`do()` superseded in dplyr 1.0.0
`reshape2::melt()` and `dcast()`	`pivot_wider()`, or dropped	The melt-dcast-clean_names round trip existed only to line two pooled rows up side by side
`geom_errorbarh(height = 0)`	`geom_linerange()`	Deprecated in ggplot2 4.0.0
`ggstance::position_dodgev()`	dropped	Still available, but inert: every y value it dodges is already unique
`ggplot() %+% data`	`ggplot(data)`	Deprecated in ggplot2 4.0.0
commented-out `ggsave()`, `print.xtable()`, `sink()`	`ggsave()` and `write_csv()` to `output/`	So that a rerun can be diffed against a committed result
`%>%`	`|>`	Native pipe

The estimator is named at every call site. `re_meta()` in `helpers.R` takes `method` as a required argument rather than defaulting to `DL`, so no script can pool without saying which estimator it pooled with. `table_2_meta_robustness.R` reports both: the DL columns

reproduce the published table, and REML columns sit beside them as an addition rather than a correction. REML moves no estimate by more than 0.001 and widens most intervals slightly, which is what a different heterogeneity estimator on 67 studies should do.

Coverage the archive states but does not deliver. The deposit's README says `meta_analyze_cates_cand_attributes.R` produces "Figure A.1, Figures A.2 through A.45, Table A.1." It contains the code for all three, with the writes commented out. The rewrite restores the 44 per-study figures, which take four seconds of the run's six. Table A.1 is a partial exception: the published table pairs each study with a prose description of its randomized attributes ("Race, ballot length"), and the deposit carries only the variable names those descriptions were written from, so `table_a1_study_attributes.csv` lists the variable names and the counts instead. The study coverage is the checkable part, and it is where the 43-against-44 discrepancy shows up.

7 Figures

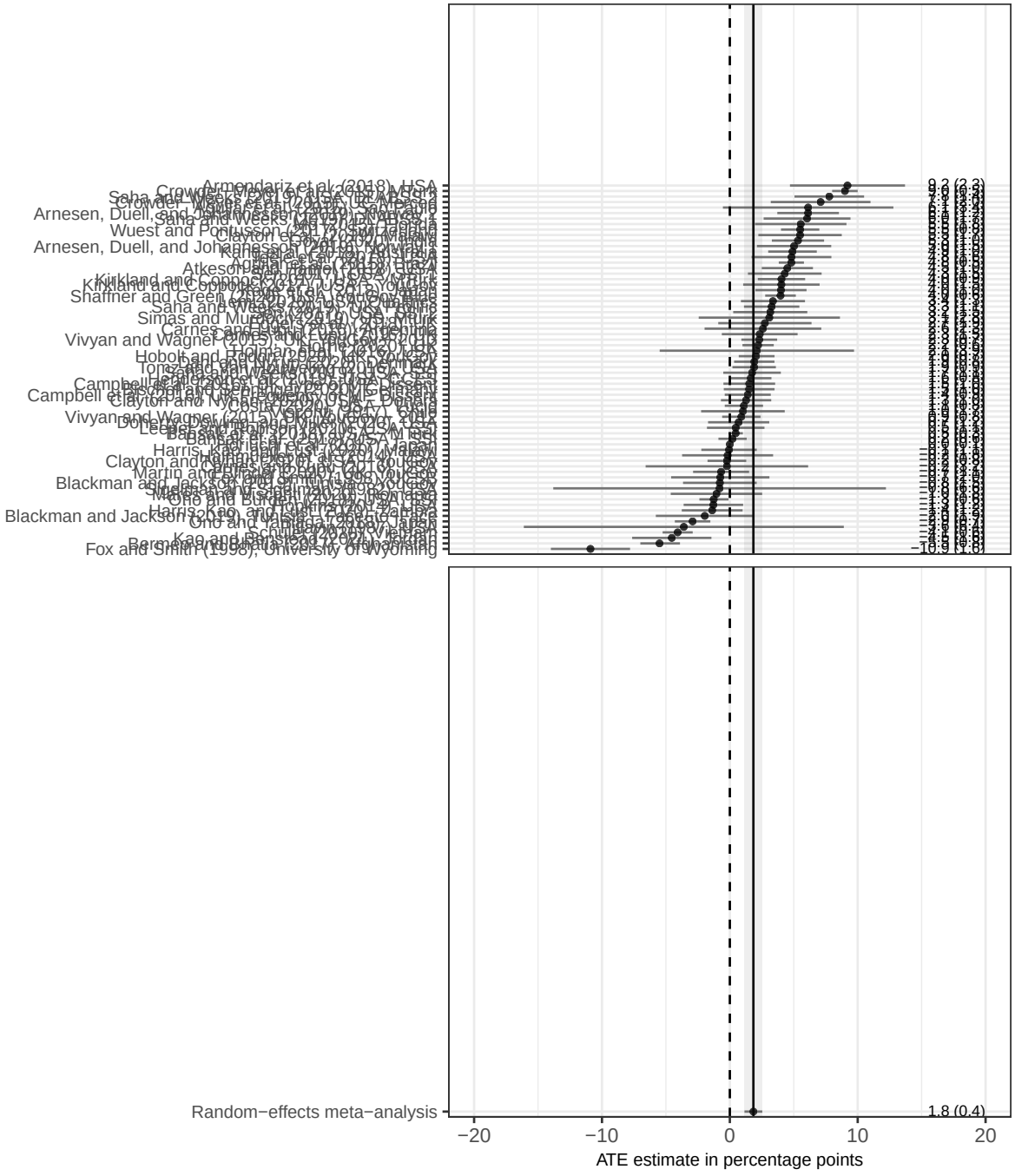


Figure 1: Figure 1. The 67 study-level ATEs and the pooled random-effects estimate.

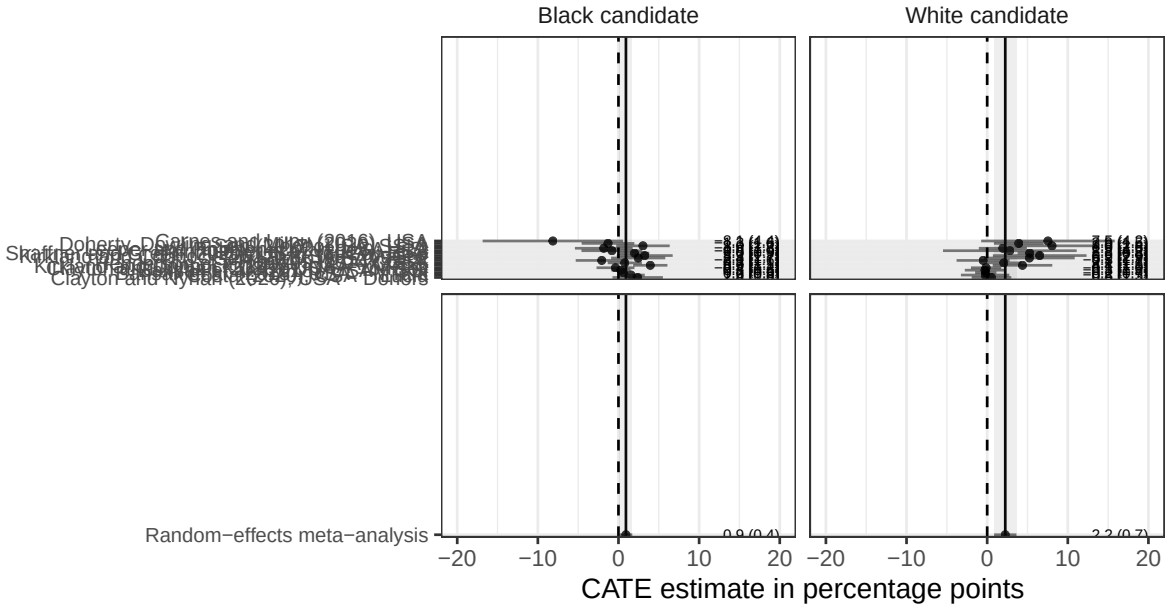


Figure 2: Figure 2. CATEs of candidate gender by candidate race.

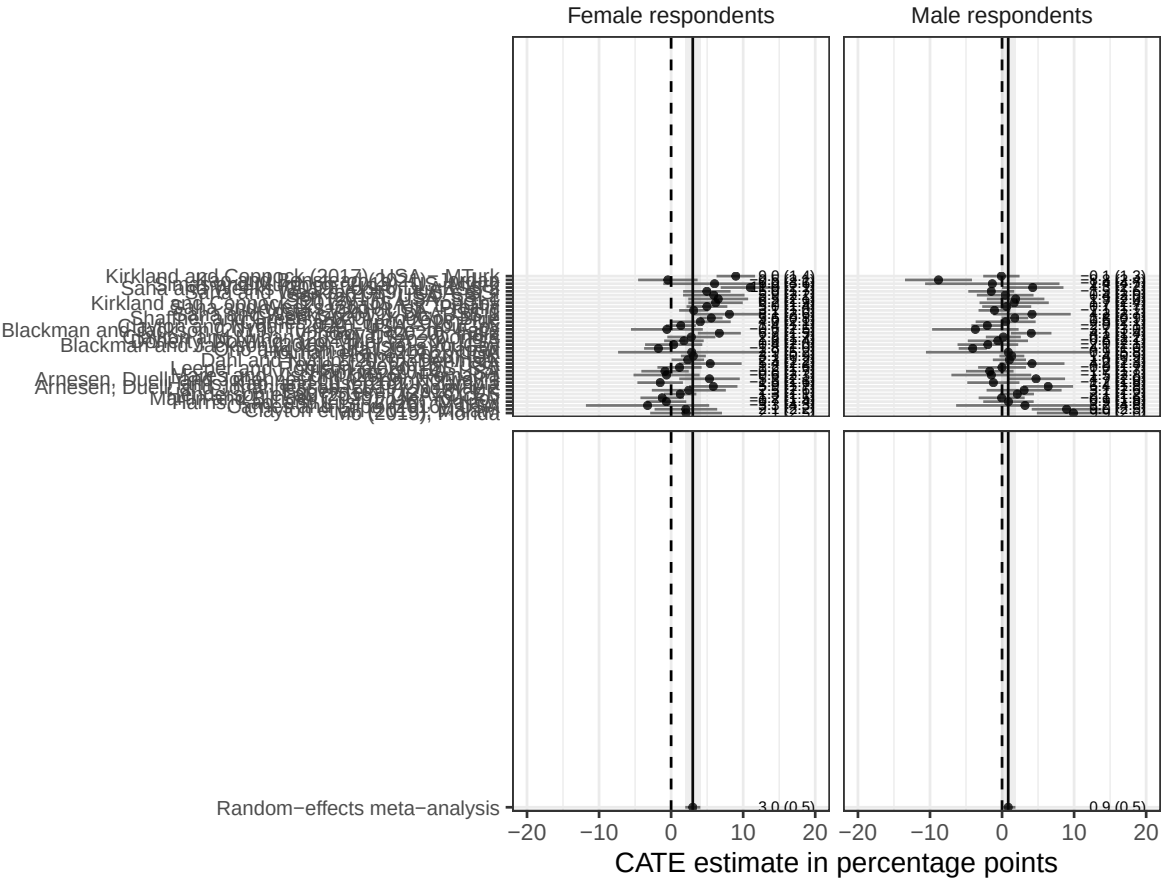


Figure 3: Figure 3. CATEs of candidate gender by respondent gender.

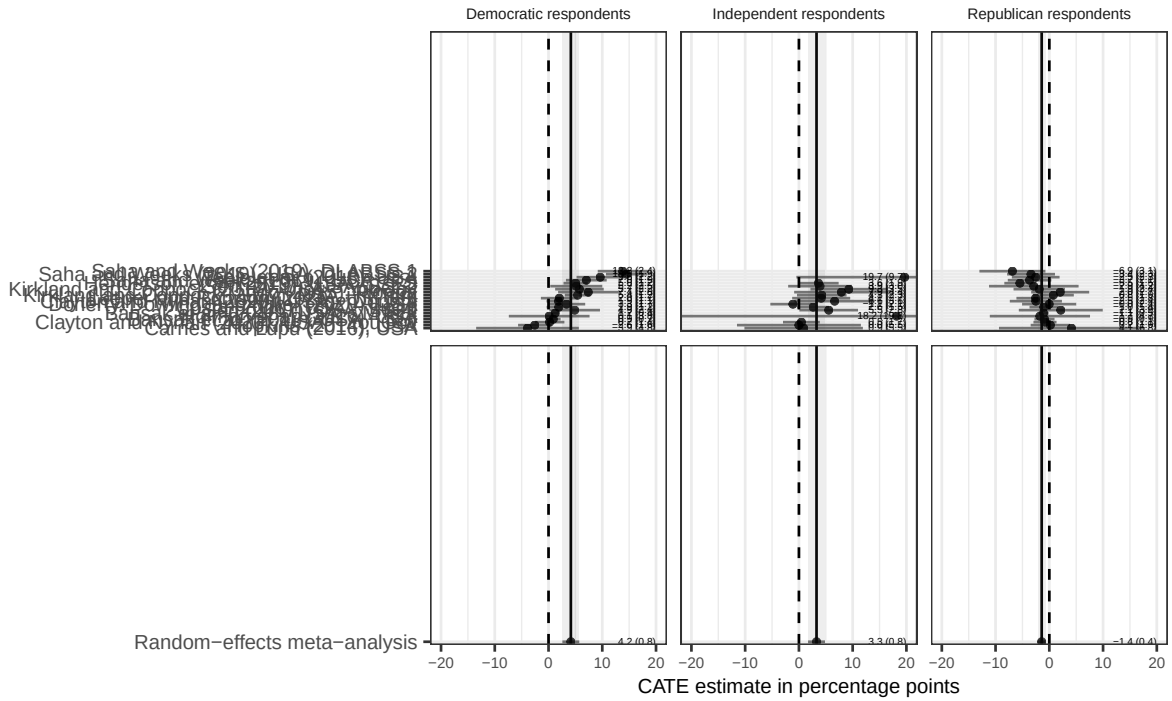


Figure 4: Figure 4. CATEs of candidate gender by respondent party identification.

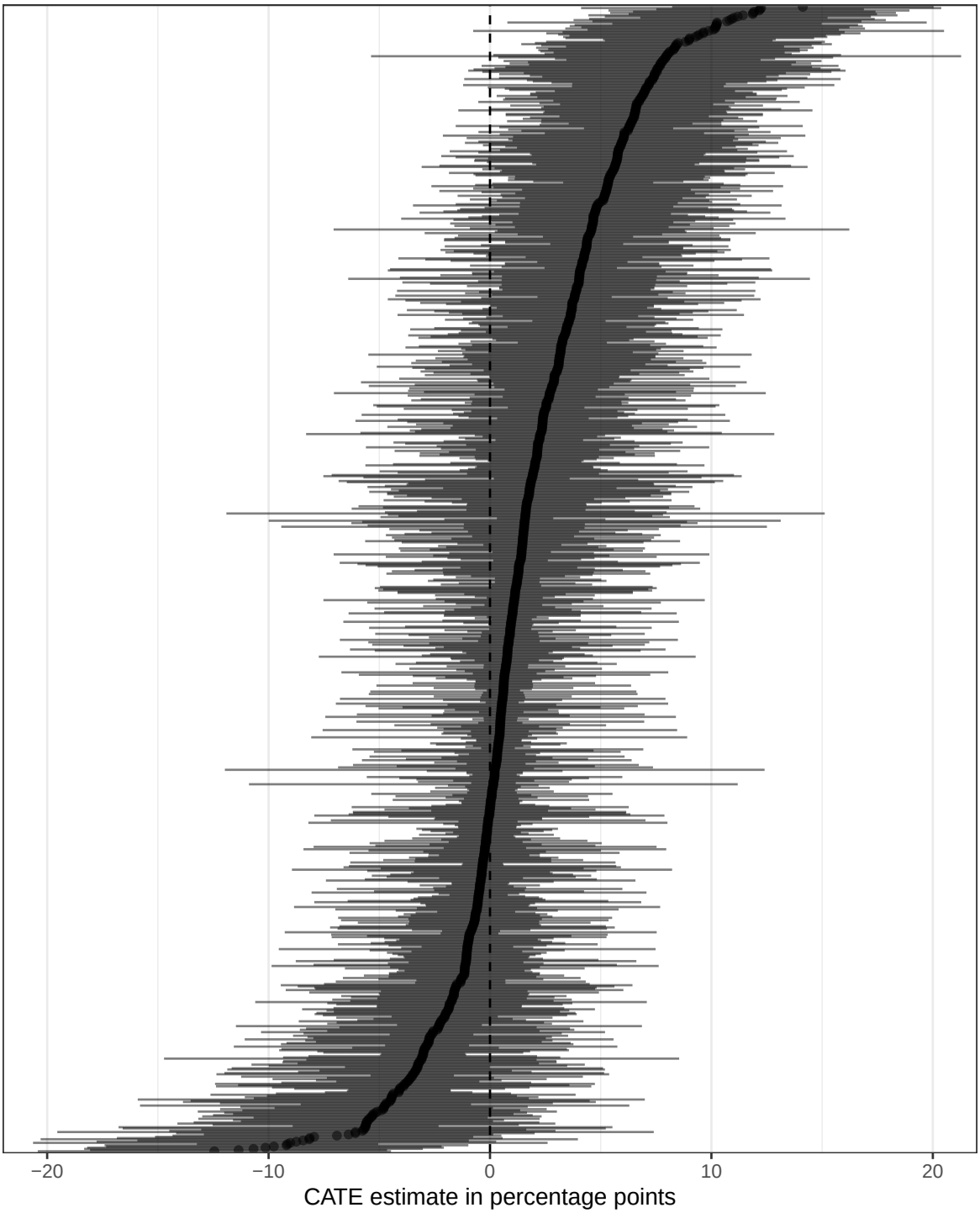


Figure 5: Figure A.1. All 954 CATEs of candidate gender conditional on the other randomized candidate attributes.

8 Rewrite verification

Table 5: Every quantity the article states, against the deposit and the rewrite (78 rows).

Table or figure	Claim	Article	Deposit	Rewrite	Deposit matches	Rewrite m
table_1	n_studies	67.000	67.000	67.000	1	
table_1	n_subjects_total	120601.000	120601.000	120601.000	1	
table_1	n_ratings_total	774971.000	774971.000	774971.000	1	
figure_1	re_meta_estimate_pp	1.800	1.842	1.842	1	
figure_1	re_meta_se_pp	0.400	0.355	0.355	1	
figure_1	re_meta_ci_low_pp	1.100	1.147	1.147	1	
figure_1	re_meta_ci_high_pp	2.500	2.537	2.537	1	
figure_1	n_studies	67.000	67.000	67.000	1	
text	n_positive	47.000	47.000	47.000	1	
text	n_negative	18.000	19.000	19.000	0	
text	n_exactly_zero	1.000	1.000	1.000	1	
text	n_pos_and_sig_ci_rule	23.000	29.000	29.000	0	
text	n_neg_and_sig_ci_rule	11.000	6.000	6.000	0	
text	prop_positive	0.750	0.701	0.701	0	
table_2	post_2014_studies_estimate	0.021	0.021	0.021	1	
table_2	post_2014_studies_std_error	0.004	0.004	0.004	1	
table_2	post_2014_studies_conf_low	0.014	0.014	0.014	1	
table_2	post_2014_studies_conf_high	0.028	0.028	0.028	1	
table_2	post_2014_studies_n_studies	63.000	63.000	63.000	1	
table_2	pre_1998_studies_estimate	-0.047	-0.047	-0.047	1	
table_2	pre_1998_studies_std_error	0.037	0.037	0.037	1	
table_2	pre_1998_studies_conf_low	-0.118	-0.118	-0.118	1	
table_2	pre_1998_studies_conf_high	0.025	0.025	0.025	1	
table_2	pre_1998_studies_n_studies	4.000	4.000	4.000	1	
table_2	convenience_student_sample_estimate	0.023	0.023	0.023	1	
table_2	convenience_student_sample_std_error	0.006	0.006	0.006	1	
table_2	convenience_student_sample_conf_low	0.012	0.012	0.012	1	
table_2	convenience_student_sample_conf_high	0.034	0.034	0.034	1	
table_2	convenience_student_sample_n_studies	31.000	31.000	31.000	1	
table_2	representative_sample_estimate	0.015	0.015	0.015	1	
table_2	representative_sample_std_error	0.005	0.005	0.005	1	
table_2	representative_sample_conf_low	0.004	0.004	0.004	1	
table_2	representative_sample_conf_high	0.025	0.025	0.025	1	
table_2	representative_sample_n_studies	36.000	36.000	36.000	1	
table_2	american_sample_estimate	0.023	0.023	0.023	1	
table_2	american_sample_std_error	0.006	0.006	0.006	1	
table_2	american_sample_conf_low	0.011	0.011	0.011	1	
table_2	american_sample_conf_high	0.035	0.035	0.035	1	
table_2	american_sample_n_studies	35.000	35.000	35.000	1	
table_2	non_american_sample_estimate	0.014	0.014	0.014	1	
table_2	non_american_sample_std_error	0.005	0.005	0.005	1	
table_2	non_american_sample_conf_low	0.004	0.004	0.004	1	
table_2	non_american_sample_conf_high	0.023	0.023	0.023	1	
table_2	non_american_sample_n_studies	32.000	32.000	32.000	1	
table_2	trimmed_middle_90_of_estimates_estimate	0.018	0.018	0.018	1	
table_2	trimmed_middle_90_of_estimates_std_error	0.003	0.003	0.003	1	

Table 5: Every quantity the article states, against the deposit and the rewrite (78 rows). *(continued)*

Table or figure	Claim	Article	Deposit	Rewrite	Deposit matches	Rewrite m
table_2	trimmed_middle_90_of_estimates_conf_low	0.012	0.012	0.012	1	
table_2	trimmed_middle_90_of_estimates_conf_high	0.023	0.023	0.023	1	
table_2	trimmed_middle_90_of_estimates_n_studies	61.000	61.000	61.000	1	
table_2	all_studies_estimate	0.018	0.018	0.018	1	
table_2	all_studies_std_error	0.004	0.004	0.004	1	
table_2	all_studies_conf_low	0.011	0.011	0.011	1	
table_2	all_studies_conf_high	0.025	0.025	0.025	1	
table_2	all_studies_n_studies	67.000	67.000	67.000	1	
figure_2	black_cand_estimate_pp	0.900	0.924	0.924	1	
figure_2	black_cand_se_pp	0.400	0.404	0.404	1	
figure_2	white_cand_estimate_pp	2.200	2.243	2.243	1	
figure_2	white_cand_se_pp	0.700	0.705	0.705	1	
figure_2	n_studies	16.000	16.000	16.000	1	
figure_2	diff_in_cates_pvalue	0.104	0.104	0.104	1	
figure_3	female_resp_estimate_pp	3.000	3.006	3.006	1	
figure_3	female_resp_se_pp	0.500	0.532	0.532	1	
figure_3	male_resp_estimate_pp	0.900	0.855	0.855	1	
figure_3	male_resp_se_pp	0.500	0.510	0.510	1	
figure_3	n_studies_in_figure	37.000	37.000	37.000	1	
figure_3	n_studies_in_text	36.000	37.000	37.000	0	
figure_3	diff_in_cates_pvalue	0.004	0.004	0.004	1	
figure_4	dem_resp_estimate_pp	4.200	4.154	4.154	1	
figure_4	dem_resp_se_pp	0.800	0.799	0.799	1	
figure_4	ind_resp_estimate_pp	3.300	3.277	3.277	1	
figure_4	ind_resp_se_pp	0.800	0.805	0.805	1	
figure_4	rep_resp_estimate_pp	-1.400	-1.418	-1.418	1	
figure_4	rep_resp_se_pp	0.400	0.363	0.363	1	
figure_4	n_studies	20.000	20.000	20.000	1	
figure_a1	n_cates	954.000	954.000	954.000	1	
figure_a1	prop_cates_positive	0.500	0.699	0.699	1	
figure_a1	n_attribute_studies	44.000	44.000	44.000	1	
table_a1	n_studies_listed	43.000	44.000	44.000	0	

The rewrite reproduces the deposit exactly. Across all 100 recorded quantities the two agree to better than 1e-10 in every case but one, the Independent-Republican p-value corrected in the Errata above. Where the deposit fails to match the article, so does the rewrite, which is the intended behaviour: the failures are in counts the article states, not in anything the code computes.

9 R environment

Table 6: Packages loaded when this report was rendered.

Package	Version
dplyr	1.2.1
forcats	1.0.1
ggplot2	4.0.3
here	1.0.2
kableExtra	1.4.0
knitr	1.51
lubridate	1.9.5
purrr	1.2.2
readr	2.2.0
stringr	1.6.0
tibble	3.3.1
tidyr	1.3.2
tidyverse	2.0.0

Rendered under R version 4.6.0 (2026-04-24) on macOS Tahoe 26.5.2.